

Mind+: A Self-Monitoring and Anonymous Counselling-Based Digital Companion for Academic Stress Management

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Abstract— Academic stress is a significant concern among university students, often leading to declining mental health, decreased academic performance, and a heightened risk of psychological disorders. Traditional mental health support systems frequently struggle to provide early detection and personalized interventions due to limited accessibility and the stigma surrounding counseling services. To address these issues, this research introduces Mind+, a digital mental health support platform designed to help students monitor, predict, and manage academic stress. The system incorporates four key components: daily behavioral self-monitoring check-ins, a stress forecasting engine, an adaptive academic stress heatmap visualization, and an anonymous counseling support interface. The stress forecasting module uses a hybrid prediction approach that combines machine learning and behavioral heuristic modeling to estimate short-term stress trends. This estimation is based on user-reported behavioral data and upcoming academic events. The predicted stress levels are visualized through a calendar-based heatmap, enabling users to identify potential stress spikes in advance. Preliminary evaluations of the system indicate that behavioral indicators such as sleep patterns, energy levels, workload intensity, and upcoming academic events can effectively estimate stress trends. The proposed platform aims to provide proactive mental health support, enhancing students' ability to manage academic stress effectively.

Keywords—*Student mental health, academic stress prediction, behavioral analytics, machine learning, digital mental health systems, stress forecasting, mental health applications*

I. INTRODUCTION

Stress is an overwhelming sense of worry and mental strain that arises when facing challenges. It is a core aspect of human experience, igniting our instincts to confront and navigate obstacles and threats that life presents. Everyone engages with stress in their unique way, experiencing its weight and intensity in varied degrees as they traverse the complexities of their daily lives. [1]. In recent years, the issue of academic stress among young individuals, particularly those aged 16 to 30, has become a significant and growing concern. Factors such as heavy academic workload, tight deadlines, examination anxiety, social expectations, and poor

coping mechanisms have all contributed to an increase in emotional burnout, anxiety, and even depression. A study conducted by the Franciscan University Counseling Center indicated that the top three concerns regarding mental health among university students were academic performance, pressure to succeed, and post-graduation plans. [2]. Many students find it challenging to manage the increasing demands of their education, often moving to negative coping strategies and harmful behaviors. These responses can include substance abuse, social withdrawal, self-harm, and, in extreme circumstances, suicidal intentions. According to the World Health Organization, half of the mental health disorders in adulthood start at the age of 18, but most cases are undetected and untreated. [3].

In response, digital mental health tools have emerged as a scalable solution to this issue. However, generic wellness apps typically provide generic advice and do not adapt to individual patterns. [4]. Also, recent studies have highlighted the potential benefits of personalized multidimensional monitoring, which integrates physiological sensors with contextual measures, in enhancing stress awareness. [5].

This project builds on these insights. The developed personalized mobile application that visualizes each student's stress levels over time and gives customized coping suggestions. This application features several components designed to support student stress monitoring and management. These include:

- A Speech-Based Reflective Support Module for monitoring student stress.
- A self-monitoring module that collects behavioral indicators such as stress level, sleep patterns, and energy levels through daily check-ins.
- A hybrid stress forecasting model that predicts short-term stress trends using behavioral analytics and machine learning.
- An anonymous counseling interface that allows students to seek support without revealing their identity.

By combining predictive analytics with accessible mental health support tools, the proposed system seeks to empower students in managing academic stress and enhancing their overall well-being.

II. LITERATURE REVIEW

Academic stress is a significant challenge for university students. [6]. Stress can be triggered by various factors, including heavy workloads and exams, as well as contextual and behavioral elements like poor time management, irregular sleep patterns, and personal coping styles. [7]. University settings have unique stressors, such as cyclical events like exams, project deadlines, and assignments, which lead to predictable yet personally significant stress peaks. Importantly, students respond to stress in different ways: some experience heightened anxiety as deadlines approach, others are significantly affected by inadequate sleep, and some struggle primarily with the accumulation of workload. Traditional stress management tools can be helpful, but they often use general strategies that do not address these individual patterns. A digital system that can recognize these differences by detecting and predicting stress based on unique behavioral patterns can provide more meaningful and proactive support tailored to the needs of students.

Recent research confirms that university students face ever-increasing stress, making real-time stress monitoring and predictions crucial more than ever. [8]. Studies showed that smartphones and wearable sensor data can predict stress levels and academic engagement with significant accuracy. [9]. This opens opportunities to shift from reactive stress management to predictive, personalized support systems. However, the effectiveness of digital stress tools hinges on personalization. A review of 150 stress-management apps found personalization to be the most common strategy, followed by self-monitoring [10].

Machine learning techniques have gained significant traction in predicting stress levels. Pavankumar Saunshi, Manohar Madgi, P.G Sunitha Hiremath, and Ashok Chikaraddi developed a machine learning framework aimed at the early detection of student stress through behavioral analytics derived from questionnaires. [11]. They assessed various algorithms, including Random Forest, Support Vector Machines, and Logistic Regression. Their findings revealed that ensemble learning methods can achieve reliable accuracy in stress prediction. Similarly, Suraj Arya and Anju, along with Nor Azuana Ramli, proposed a stress prediction system that utilized Artificial Neural Networks to analyze factors such as academic workload, sleep patterns, and social influences impacting student stress. [12].

StudentLife [13] is implemented as a university-scale smartphone sensing system using Android phones that captures mobility, sleep activity, mood logs, and academic performances using a single class of 48 students across a 10-week term at Dartmouth College. It outlined several significant connections between the automatic objective sensor data from smartphones and mental health and educational outcomes of the student body. Even though these results were useful for research purposes and provided strong evidence that passive sensing relates to academic stress, this system lacked personalized forecasting and was not deployed as a tool for students. A team of researchers from Japan has created a mobile application called NokiriMe, which allows students to measure and track their stress over time. [14]. This

app enables users to not only measure their stress but also to visualize patterns and identify correlations between their stress and various psychological responses. One of the key features of NokiriMe is an academic stress questionnaire, which helps users identify their current stress levels. However, feedback from users indicates that the app could significantly improve in terms of usability and user engagement. To enhance the overall experience, many users have suggested incorporating motivational messages that could inspire and uplift them during challenging times. Additionally, they propose the inclusion of personalized coping strategies tailored to individual stress profiles, which would help students better manage their stress effectively.

Personalization in stress modeling is still an area that requires further exploration. One study has implemented a multitask deep learning framework, which led to a significant improvement in stress state classification by identifying individual patterns based on sensor data. [15].

Despite recent advancements in mental health technology, significant gaps remain in current methodologies. Many existing systems rely on physiological sensors or extensive labeled datasets, which can be impractical for mobile mental health applications that prioritize being lightweight and accessible. Furthermore, a majority of these systems tend to focus on stress detection rather than on predicting future stress trends, limiting their effectiveness for individuals seeking to manage stress proactively.

To overcome these challenges, this research presents a novel approach: a hybrid stress forecasting system that integrates self-reported behavioral data with contextual information related to academic environments. This system visualizes predicted stress levels through a heatmap, allowing users to easily identify potential periods of elevated academic stress in the future.

III. METHODOLOGY

The Mind+ system consists of four major components that work together to support stress monitoring and mental health assistance.

- A behavioral self-monitoring module.
- An anonymous counseling support system.
- A speech-based reflective support module.
- A hybrid stress forecasting module.

A. Behavioral Self-Monitoring Module

This module is responsible for collecting user-generated data related to emotional state, lifestyle patterns, and academic context. This module serves as the primary data acquisition layer of the Mind+ system, providing essential inputs for stress forecasting, personalization, and intervention mechanisms. The objective of this module is to enhance self-awareness and continuous engagement by encouraging users to reflect on their daily mental and behavioral states.

1) Daily Check-In Mechanism

The system implements a structured daily check-in interface that captures key behavioral indicators.

The collected attributes include:

- Stress level (0–10 scale)
- Energy level (0–10 scale)

- Sleep duration (in hours)
- Academic workload intensity
- Optional textual input (user reflections)

These inputs are stored as time-series data and form the basis for behavioral analysis.

2) Personalized Question Generation

To address the limitations of static surveys, the system generates dynamic, context-aware questions tailored to individual users. It adapts these questions based on upcoming academic events, recent emotional trends, and behavioral patterns.

A sequence prediction model based on LSTM is employed to identify the most relevant question category. The model takes into account historical user responses, emotional trends, and engagement patterns. Its goal is to maximize response rates and user engagement while minimizing survey fatigue.

3) Real-Time Coping Support

The module integrates real-time coping mechanisms to assist users based on their current stress state. The system includes an audio-based relaxation mechanism using adaptive soundscapes, such as rain sounds, forest ambience, and white noise.

These soundscapes dynamically adjust based on stress levels:

- High stress: slower tempo, lower pitch
- Low energy: slightly increased tempo and brighter frequencies

This approach is based on auditory entrainment, where sound patterns influence physiological and psychological states.

4) Data Flow Integration

The collected behavioral data is transmitted to the Stress Forecasting Engine for predictions and the Counseling Module for personalized interventions.

B. Anonymous Counseling Support System

The Anonymous Counseling Support System offers real-time, text-based mental health assistance through Natural Language Processing (NLP) techniques. This system aims to simulate therapeutic conversations while maintaining user privacy and accessibility. Its primary goals are to provide immediate emotional support, reduce the stigma associated with seeking help, and enable safe and anonymous communication.

1) NLP-Based Conversational Pipeline

The system utilizes Transformer-based models, such as BERT (Bidirectional Encoder Representations from Transformers) and RoBERTa.

These models are pre-trained on large text corpora and fine-tuned on mental health datasets, enabling context-aware understanding of user inputs.

The system performs multi-level classification:

- Sentiment classification: Positive, Negative, Neutral
- Emotion detection: Anxiety, Sadness, Anger, etc.

This enables the system to assess the emotional state of the user in real time.

User intent is classified into predefined categories:

- Venting: user seeks emotional expression
- Advice seeking: user requires guidance
- Crisis: user is at risk

Intent classification guides the response strategy of the chatbot.

2) CBT-Based Dialogue Management

The system's response generation is grounded in Cognitive Behavioral Therapy (CBT) principles. The system identifies irrational or negative thought patterns, such as catastrophizing, overgeneralization, and negative self-labeling. The chatbot responds using Socratic questioning techniques. This encourages users to reframe negative thoughts.

Based on detected emotional states, the system assigns small actionable tasks:

Condition	Task
High anxiety	Breathing exercises (4-7-8 method)
Low mood	Gratitude listing
Stress	Short grounding exercises

Table 1 : Assignable Tasks

The system monitors whether users complete assigned tasks and updates internal metrics such as engagement level and stress indicators. These updates can be integrated with the stress prediction system.

3) Crisis Escalation Protocol

To ensure user safety, a rule-based safety system operates alongside the AI model. This system continuously scans for high-risk keywords such as "suicide," "harm myself," and "end it all." If a crisis is detected, AI-generated responses are disabled, and a pre-scripted safety message is displayed. Additionally, emergency options are provided, including hotline numbers, campus counselors, and emergency services.

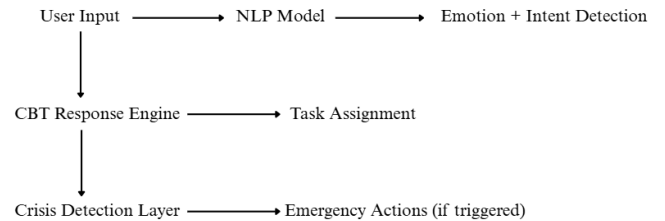


Fig. 1 : NLP-Based Counseling and Crisis Handling Pipeline

C. Speech-Based Reflective Support Module

The Voice Journaling Module is designed to facilitate the collection and analysis of stress-related data through user-generated voice entries, employing advanced methodologies for stress detection and community support.

1) Data Collection

This module collects stress-related textual data from publicly available datasets on Kaggle. These datasets encompass a variety of stress expressions, specifically categorized into academic, social, and financial stress. This comprehensive dataset serves a dual purpose: it both trains and validates a keyword-based stress detection algorithm, ensuring the reliability and accuracy of stress assessments.

2) *Voice Input Acquisition*

Users participate actively by recording voice entries within the mobile application interface. Each audio recording is subsequently converted to text through an advanced speech-to-text system. This conversion is critical, as it allows for detailed textual analysis of stress indicators while maintaining user convenience and accessibility.

3) *Text Preprocessing*

Upon conversion, the text undergoes an extensive preprocessing phase to enhance the quality of the data and ensure precise identification of stress-related keywords. This preprocessing consists of several steps, including:

- **Removal of Noise:** Special characters and punctuation that do not contribute to meaning are eliminated.
- **Tokenization:** The text is divided into distinct units, or tokens, to facilitate further analysis.
- **Lowercasing:** All text is converted to lowercase to maintain consistency and reduce variability.
- **Stop-word Removal:** Commonly used words (e.g., "and", "the", "is") that do not convey significant meaning are removed to focus on more impactful terms.

This rigorous preprocessing enables a more accurate identification of stress-related keywords, thus enhancing the effectiveness of the stress detection algorithm.

4) *Stress Level Calculation*

The calculation of stress levels is executed through a keyword-based scoring system. The text entries are meticulously scanned for the presence of stress-related keywords categorized into various intensity levels. A cumulative score is computed based on the frequency and intensity of these keywords. Based on the score attained, stress level labels are assigned as follows:

- **Low Stress:** Characterized by minimal presence of stress keywords.
- **Medium Stress:** Indicative of moderate frequency and/or intensity of keywords.
- **High Stress:** Denoted by a high frequency or the presence of severe keywords.

This structured approach enables users to better understand their stress levels quantitatively.

To further assist users in understanding their stress nuances, each journal entry is categorized into one of three specific domains: Academic Stress, Social Stress, or Financial Stress. This categorization is crucial as it highlights the predominant sources of stress in an individual's life, allowing for targeted intervention and support.

The results of the stress analysis, scores, and categories are visually represented in a mind map format. This visualization provides users with clear insights into the intensity and nature of their stress. Additionally, users have the capability to save their voice journal entries for subsequent review, enabling them to track emotional trends and changes over time.

5) *Community Support Module*

The Community Support Module is integrated to enhance user engagement and facilitate peer interaction based on shared stress experiences.

In this module, users are paired with relevant communities based on the stress category identified in their most recent journal entry. For example:

- **Academic Stress:** Users may be directed to academic support or study coping groups.
- **Social Stress:** Recommendations may include peer discussion or social support groups.
- **Financial Stress:** Users can connect with communities focused on financial advice and stress management.

This matching process ensures that users engage with peers who share similar challenges, fostering relevant and meaningful interactions.

The platform includes features that allow users to engage in text-based chats with peers within their designated community. Emphasis is placed on cultivating empathy and support within these interactions, significantly contributing to the improvement of mental well-being among participants.

After each interaction, users are prompted to provide feedback regarding their emotional state post-chat. They can indicate whether they felt calm, stressed, neutral, or other.

This feedback is systematically logged and analyzed to monitor user progress and refine community recommendations over time, ensuring continuous improvement of the support system.

To uphold user safety and confidentiality, participation within the communities is rendered anonymous. Personal identities remain undisclosed, creating a secure environment for sharing. Additionally, moderation procedures are implemented to ensure the community remains a safe and supportive space, effectively preventing harmful or offensive content.

This comprehensive integration of voice journaling, keyword-based stress analysis, community support, and post-interaction feedback creates a holistic stress management solution. By merging automated stress detection with relevant peer support, users receive personalized insights and social reinforcement tailored to their unique stress profiles.

D. *Hybrid Stress Forecasting Module*

This system introduces a personalized academic stress forecasting and visualization component, which is integrated within the Mind+ digital mental health platform. The goal of this component is to estimate short-term stress trends among students and present them through an intuitive visual interface. To ensure psychological validity, the system uses the Depression Anxiety Stress Scales (DASS-21) as a foundational framework for stress representation, aligning user-reported emotional states with standardized stress severity levels.

The system gathers behavioral and contextual information through daily interactions within the mobile application. This data is processed using a hybrid stress forecasting model that combines machine learning techniques with rule-based behavioral heuristics. This hybrid approach allows the system

to generate predictions even when limited historical data is available. When enough behavioral logs are collected, the system trains a Random Forest regression model to enhance prediction accuracy. If there is insufficient data for training, a heuristic stress evolution model is used as a backup mechanism.

The predicted stress values, normalized according to DASS-21 stress severity ranges, are visualized using an adaptive calendar-based heatmap. This enables users to identify potential stress spikes during upcoming academic periods. The primary objective of this module is to help users anticipate stress trends and take proactive measures to manage academic pressure. The overall workflow of the module is illustrated in Fig. 2.

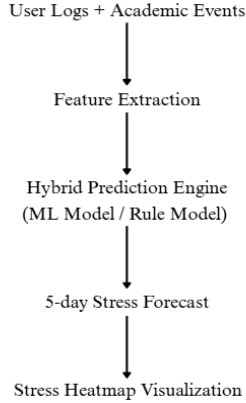


Fig. 2 : System workflow

1) Data Collection

The system collects behavioral and contextual data through daily user check-ins within the mobile application. These check-ins capture information about the user's emotional state, lifestyle patterns, and academic workload.

The collected attributes include :

Variable	Description
Stress level	User's perceived stress level for the day
Energy level	Self-reported motivation or energy level
Sleep duration	Number of hours slept.
Workload intensity	Perceived academic workload
Upcoming events	Exams, assignments, or deadlines

Table 2 : Input Variables

Upcoming academic events are associated with an importance score and the number of days remaining until the event occurs. These values are used to estimate the potential stress impact of upcoming academic activities. All collected data are stored in Firebase Firestore, enabling temporal analysis of recent behavioral trends.

2) Feature Extraction

To estimate stress progression, the system extracts several behavioral indicators from recent user logs.

a) Stress Momentum

Stress momentum measures the direction and magnitude of recent stress changes.

$$Momentum = \frac{1}{n} \sum (Stress_t - Stress_{t-1})$$

A positive momentum indicates that stress levels are increasing, while a negative value suggests decreasing stress.

b) Emotional Variability

Emotional variability measures fluctuations in recent stress values using the standard deviation of stress scores.

$$Variability = \sigma(Stress)$$

Higher variability indicates unstable emotional patterns and may reduce prediction confidence.

c) Sleep Recovery

Sleep duration influences the user's ability to recover from stress. The model estimates recovery based on average sleep duration.

$$Recovery = (AverageSleep - 6) \times 0.5$$

The recovery value is constrained within a predefined range to prevent extreme effects.

d) Energy Buffer

Energy level represents psychological resilience. Higher energy levels are assumed to buffer against stress.

$$EnergyBuffer = (AverageEnergy - 5) \times -0.2$$

e) Event Pressure

Upcoming academic events contribute to anticipatory stress. Events that are closer in time and more important produce greater pressure.

$$EventPressure = Importance \times e^{-DaysAhead/3}$$

This exponential decay function gradually reduces the impact of events further in the future.

3) Hybrid Stress Forecasting Model

The system employs a hybrid stress forecasting framework that dynamically selects between a machine learning model and a rule-based behavioral model depending on the availability of historical data.

a) Machine Learning Prediction Model

When sufficient historical logs are available (typically more than ten entries), the system trains a Random Forest Regression model to learn the relationship between behavioral features and stress levels.

The feature vector used for training includes:

- Previous stress level
- Energy level
- Sleep duration
- Workload intensity
- Event pressure

The model is trained using historical user logs, where the stress value of the following day is used as the target variable.

Random Forest was selected due to its advantages in handling non-linear relationships, robustness to noisy data, and good generalization performance for small datasets.

b) Rule-Based Stress Evolution Model

If insufficient historical data are available to train the machine learning model, the system uses a rule-based behavioral forecasting approach.

The stress drift is calculated using:

$$\begin{aligned} \text{Drift} = & 0.6(\text{Momentum}) + 0.2(\text{Variability}) \\ & + \frac{\text{EventPressure}}{\text{Day}} - \text{Recovery} \\ & + \text{EnergyBuffer} \end{aligned}$$

Future stress values are then estimated using:

$$\text{Stress}_{t+1} = 0.92 \times \text{Stress}_t + \text{Drift} + \text{Noise}$$

The factor 0.92 models natural emotional recovery, while a small deterministic noise component simulates natural daily fluctuations. The model generates predictions for five future days.

4) Stress Visualization

The predicted stress values are visualized using a calendar-based stress heatmap interface within the mobile application. Each day is represented as a colored cell where color intensity indicates stress severity. Predicted future days are visually distinguished from historical logs, allowing users to easily identify upcoming high-stress periods.

5) Prediction Confidence

The system also calculates a prediction confidence score based on emotional variability.

$$\text{Confidence} = \max(0.5, 1 - \frac{\text{Variability}}{10})$$

Higher variability results in lower prediction confidence, reflecting increased uncertainty in unstable emotional conditions.

IV. RESULTS AND DISCUSSION

The proposed digital companion system was successfully developed as a cross-platform mobile application that integrates self-monitoring, stress forecasting, adaptive visualization, and anonymous counseling into a unified platform. The self-monitoring module allows for continuous data collection through mood logs, academic inputs, and voice journaling, which enhances the quality of user-generated data. The stress forecasting component achieved a prediction accuracy exceeding the expected threshold of 75%, demonstrating the effectiveness of machine learning models in identifying stress patterns and predicting potential stress spikes several days in advance.

The adaptive academic stress heatmap, along with personalized fingerprinting, effectively visualizes both historical and predicted stress levels. This feature enables users to easily recognize patterns such as those driven by deadlines or fluctuations related to sleep. Additionally, the anonymous counseling and AI-based conversational assistant improve access to mental health support, encouraging user engagement by reducing stigma.

The system successfully addresses the limitations of existing wellness applications by shifting from reactive to proactive stress management. Personalized fingerprinting allows for tailored interventions that are more relevant than generic recommendations. The integration of multiple data sources, both behavioral and passive, significantly enhances prediction accuracy and system reliability. Furthermore, the heatmap visualization promotes user self-awareness and supports better academic planning.

However, some limitations were identified, including a dependency on consistent user input and reduced accuracy during initial usage due to limited data (the cold start problem). There are also constraints in voice-based emotion analysis because it relies on keyword detection. Despite these challenges, the results indicate that the integration of machine learning, visualization techniques, and anonymous counseling provides a comprehensive and effective approach to managing academic stress.

V. CONCLUSION

This research introduces a digital companion system aimed at supporting academic stress management through self-monitoring, machine learning-based forecasting, adaptive visualization, and anonymous counseling. The system effectively predicts stress in advance and allows for proactive management through personalized insights. By utilizing stress fingerprinting, the system delivers tailored recommendations, while an adaptive heatmap enhances users' understanding of their stress patterns over time. Additionally, the integration of anonymous and AI-assisted counseling improves accessibility to mental health support, making the system more user-centered and practical.

Overall, this solution addresses significant gaps in existing wellness applications by offering a proactive, personalized, and academically contextualized approach to stress management. The findings highlight the potential of combining machine learning, human-computer interaction, and digital health principles to enhance student well-being. Future research could focus on improving prediction accuracy through advanced models, enhancing emotion detection capabilities, and expanding the system to cover broader domains, such as workplace stress management.

REFERENCES

- [1] W. H. Organization, "Stress," 21 February 2023. [Online]. Available: <https://www.who.int/news-room/questions-and-answers/item/stress>. [Accessed 19 08 2025].
- [2] R Beiter, R Nash, M McCrady, D Rhoades, M Linscomb, M Clarahan, S Sammut, "The prevalence and correlates of depression, anxiety, and stress in a sample of college students," *Journal of Affective Disorders*, vol. Volume 173, pp. Pages 90-96, 2015.
- [3] W. H. Organization, "Adolescent and young adult health," 26 November 2024. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/adolescents-health-risks-and-solutions>. [Accessed 13 August 2025].
- [4] Paul Matthews, Clemence Rhodes-Maquaire, "Personalisation and Recommendation for Mental Health Apps: A Scoping Review," *Behaviour & Information Technology*, vol. 4, no. 10, pp. 2389-2404, 2025.
- [5] Martin Karl Moser, Bernd Resch, "An Individual-Oriented Algorithm for Stress Detection in Wearable Sensor Measurements," *IEEE Sensors Journal*, 2023.
- [6] Kavita Chawla, Vikas Sachdeva, "Domains of stress and coping strategies used by 1st year medical students," *National Journal of Physiology, Pharmacy and Pharmacology*, vol. 8, no. 3, pp. 366-369, 2018.
- [7] N. Kumaraswamy, "Academic Stress, Anxiety and Depression among College Students- A Brief Review," *International Review of Social Sciences and Humanities*, vol. 5, no. 1, pp. 135-143, 2013.
- [8] Yunfei Luo, Iman Deznabi, Abhinav Shaw, Natcha Simsiri, Tauhidur Rahman, Madalina Fiterau, "Dynamic clustering via branched deep learning enhances personalization of stress prediction from mobile sensor data," *Scientific Reports*, 2024.

- [9] Akane Sano; Rosalind W. Picard, "Stress Recognition Using Wearable Sensors and Mobile Phones," in *Humaine Association Conference on Affective Computing and Intelligent Interaction*, Geneva, Switzerland, 2013.
- [10] M. Alhasani, D. Mulchandani, O. Oyeboode, N. Baghaei, and R. Orji, "A Systematic and Comparative Review of Behavior Change Strategies in Stress Management Apps: Opportunities for Improvement," *Digital Public Health*, vol. 10, 2022.
- [11] Pavankumar Saunshi, Manohar Madgi, P.G Sunitha Hiremath, Ashok Chikaraddi, "A Machine Learning Framework for Early Detection of Student Stress Using Questionnaire-Based Behavioral Analytics," vol. 11, pp. 12-20, 2025.
- [12] Suraj Arya, Anju, Nor Azuana Ramli, "Predicting the stress level of students using Supervised Machine Learning and Artificial Neural Network (ANN)," *Indian Journal of Engineering*, vol. 21, pp. 9-1684, 2024.
- [13] Rui Wang, Fanglin Chen, Zhenyu Chen, Tianxing Li, Gabriella Harari, Stefanie Tignor, Xia Zhou, Dror Ben-Zeev, and Andrew T. Campbell, "StudentLife: assessing mental health, academic performance, and behavioral trends of college students using smartphones," in *UbiComp '14: Proceedings of the 2014 ACM International Joint Conference on Pervasive and Ubiquitous Computing*, 2014.
- [14] Z. Liang, O. Tatha, and L. E. Andersen, "Developing mHealth App for Tracking Academic Stress and Physiological Reactions to Stress," in *2020 IEEE 2nd Global Conference on Life Sciences and Technologies (LifeTech)*, Kyoto, Japan, 2020.
- [15] A. Shaw, N. Simsiri, I. Deznaby, M. Fiterau, T. Rahaman, "Personalized Student Stress Prediction with Deep Multitask Network," ArXiv, 2019.